

Part 1: Agrivoltaics Requirements Document

This is a requirements document for an agrivoltaics research platform. The array will be used for collaborative research on dual use agrivoltaics systems with an interdisciplinary and inter-sector network of research partners to study crop performance, soil health, environmental impact and economic benefits. The array will also be used for extensive education and demonstration to diverse audiences about agrivoltaics as a solution to food and energy security and farmland preservation. The design is intended to provide flexibility to test a variety of crops, farming equipment and practices, that are mainly relevant to farmers in Maryland and mid-Atlantic states. Note that only agrivoltaics specific considerations are included in this documents and it is not exhaustive of all PV design specifications or other engineering specifications.

Design Considerations

Rough, proposed layout schematic:



Four Proposed configurations are each approximately 500 kW capacity

1. Single Axis Tracker (SAT) 6 ft hub height, ground coverage ratio (GCR): 0.37
SAT rows run north-south but some consideration should be included for slight deviation to follow contours of plot
2. SAT, 8 ft hub height, GCR: 0.37
3. SAT, 6 ft hub height, GCR: 0.21
4. Fixed – south, tilt: 35 deg, 8 ft mounting height, GCR: 0.37

*tilt capability of ~60 degrees is preferred for agricultural clearance and flexibility for shading experiments

- System should be able to accommodate sensors for monitoring microclimate, soil conditions and vegetation growth performance
- Bifacial panels are preferred
- Tracking systems that include ability to override automated tilting are preferred to allow for more flexibility to control shading for experimental purposes and adjust at anytime.
- Use ground screws or similar low impact anchor and do not use above grade footers. Subsurface foundations should avoid long term soil disruption.
- Conduit runs and system components should be located close to racking footings to avoid interference with agricultural operations
- Wiring routes must avoid crop zones and be clearly marked in as-built documentation
- Field and edge buffers should be included and coordinated with farming partner (starting point of 25 ft for tractor turning radius). Include north-south farming access pathways between each of the four proposed PV areas
- Inverter and junction box should be staged outside PV array to allow for farming equipment access
- Wire management practices should be included to avoid loose wiring and provide safety to farmers inside array. No array components (above or below ground) should interfere with agricultural activities (excluding the array posts)
 - DC cables should be installed high enough to avoid human and/or livestock interaction and AC cables at least 3 ft below ground (for tilling/digging operations in surrounding farm area)

Construction Considerations

- Minimize disturbance (grading, cutting, filling, trenching) and compaction (through providing staging and driving areas for heavy equipment) in surrounding area where in-soil farming will occur
- Where necessary for soil removal, topsoil must be preserved and returned as surface layer
- Requirements for well need to be coordinated with farming and/or research partners prior to construction